



ABES Engineering College, Ghaziabad
Department of Applied Sciences & Humanities

Session: 2025-26

Semester: I

Course Code: 25AS102

Course Name: Fundamentals of Linear Algebra & Statistics

Max. Marks:10

Assignment 4

Date of Assignment:

Date of submission:

S.No.	KL	CO	Question	Marks																		
1	K3	CO4	The weekly sales of two products A and B were recorded as given below :	2																		
			<table><tr><td>A</td><td>59</td><td>75</td><td>27</td><td>63</td><td>27</td><td>28</td><td>56</td></tr><tr><td>B</td><td>150</td><td>200</td><td>125</td><td>310</td><td>330</td><td>250</td><td>225</td></tr></table>		A	59	75	27	63	27	28	56	B	150	200	125	310	330	250	225		
			A		59	75	27	63	27	28	56											
B	150	200	125	310	330	250	225															
Find out which of two shows greater fluctuations in sales.																						
2.	K3	CO4	The first four moments of the distribution about the value 4 are -1.5, 17, -30, 80. Find the moments about the mean and discuss the nature of the distribution.	2																		
3.	K3	CO4	Find the median, lower and upper quartile from the following data:	2																		
			<table><tr><td>Marks</td><td>Below 10</td><td>Below 20</td><td>Below 30</td><td>Below 40</td><td>Below 50</td><td>Below 60</td><td>Below 70</td><td>Below 80</td></tr><tr><td>No. of students</td><td>15</td><td>35</td><td>60</td><td>84</td><td>94</td><td>127</td><td>198</td><td>249</td></tr></table>		Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60	Below 70	Below 80	No. of students	15	35	60	84	94	127	198	249
			Marks		Below 10	Below 20	Below 30	Below 40	Below 50	Below 60	Below 70	Below 80										
No. of students	15	35	60	84	94	127	198	249														
4	K3	CO4	Calculate first four moments from the following frequency distribution table and hence find coefficient of skewness and kurtosis.	2																		
			<table><tr><td>x</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>f</td><td>10</td><td>20</td><td>40</td><td>20</td><td>10</td></tr></table>		x	0-10	10-20	20-30	30-40	40-50	f	10	20	40	20	10						
			x		0-10	10-20	20-30	30-40	40-50													
f	10	20	40	20	10																	

5	K3	CO4	Find the mode of the following frequency distribution							2
			Size	10-25	25-40	40-55	55-70	70-85	85-100	
			frequency	5	21	21	8	25	20	

Answers:

1. Product A

2. $\mu_1 = 0, \mu_2 = 120, \mu_3 = 0, \mu_4 = 36000$

3. Median=59.24, lower quartile=30.94, upper quartile=68.41

4. $\mu_1 = 0, \mu_2 = 14.75, \mu_3 = 39.75, \mu_4 = 114.3125, \beta_1 = 0.492377, \beta_2 = 0.52542$

5. Mode = 81.59